Lecture 27

Solitons and topological charge

Objectives. *Lecture 26 classified maps into an order-parameter space. A field theory also assigns an energy to each configuration. We separate the boundary conditions that define a topological sector from the field equations that select stationary configurations. The kink supplies a complete energy-bound calculation. The gauge vortex shows how a phase winding becomes a quantized magnetic flux.*

By the end of this lecture you will be able to:

- Identify a sector from its stated continuous vacuum boundary data.
- Derive the kink bound and its saturating profile with the signed charge.
- Prove Derrick's obstruction for the stated linear-target scalar energy, including $d = 2$.
- Derive vortex flux quantization under the stated asymptotic conditions.

27.1 Vacuum boundary data and sectors

Recall that a classical field configuration assigns a value to each spatial point. For a scalar field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, take a smooth potential $V \geq 0$. Its zero set $\mathcal{V} = V^{-1}(0)$ is the *vacuum manifold* when it is a manifold. A constant field in $\mathcal{V}$ has zero potential and gradient energy. A *static soliton* is a spatially localized, finite-energy stationary solution of the field equations. The boundary conditions below specify the allowed behaviour far from its localized region. A configuration carrying a nonzero topological charge is not automatically such a solution.

In one dimension, assume limits ϕ_- and ϕ_+ in $\mathcal{V}$ at $-\infty$ and $+\infty$. Call their path components $\mathcal{V}_-$ and $\mathcal{V}_+$. In dimensions two and three, assume a continuous angular limit $\phi_\infty\colon S^{d-1} \to \mathcal{V}$, with convergence uniform in direction. A *soliton sector* fixes the corresponding boundary homotopy data. For an evolving field, require that its limiting boundary map be jointly continuous in angle and time. In one dimension require each endpoint limit to be continuous in time. These are boundary assumptions. Finite energy alone does not imply them. The figure preserves the order of the one-dimensional endpoints.



Figure 27.1: The left and right vacuum components form an ordered pair. Interchanging the two ends can reverse a kink's charge.

Proposition 27.1 (Boundary sectors · *proved*)**.** *Under these boundary conditions, the one-dimensional datum is the ordered pair of path components of $\mathcal{V}$ containing ϕ_- and ϕ_+. In dimensions two and three it is the free homotopy class of ϕ_∞. An evolution with jointly continuous vacuum boundary data preserves these data.*

Proof. At either end in one dimension, the time-dependent limiting value is a path in $\mathcal{V}$. It cannot leave its path component. The two ends remain distinguished, giving an ordered pair. For $d = 2$ or $d = 3$, the jointly continuous boundary map is exactly a homotopy of maps $S^{d-1} \to \mathcal{V}$. Its free homotopy class is therefore fixed. Lecture 26 expresses these classes using conjugacy or the fundamental-group action when needed. ∎

A *topological charge* is an invariant used to label such a sector. An integer charge exists in the examples below, but the general sector datum need not be an integer. It constrains allowed deformations. Energy and the field equations determine whether a soliton exists.

27.2 The scalar kink and the Bogomolny bound

Take one real field on the line and constants $\mu, v > 0$. Define the superpotential W, potential V and static energy by

$$W(\phi) = \mu\left(v^2\phi - \frac{\phi^3}{3}\right), \qquad V(\phi) = \frac{1}{2}\big(W'(\phi)\big)^2, \qquad E = \int_{\mathbb{R}} \left(\frac{1}{2}(\phi')^2 + V(\phi)\right) \mathrm{d}x.$$

Here W' differentiates with respect to the field value, while ϕ' differentiates with respect to x. The vacua are $-v$ and $+v$. Assume the specified vacuum limits at both ends. The figure shows the increasing kink profile centred at zero.

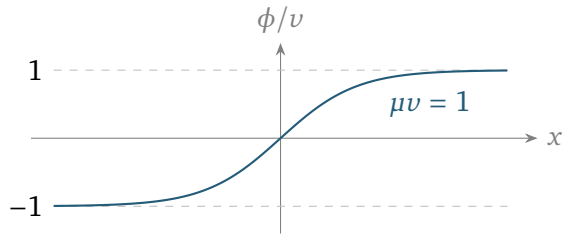


Figure 27.2: The exact kink $\phi/v = \tanh(\mu v x)$ joins $-v$ to $+v$. The increasing profile has charge +1.

Proposition 27.2 (The kink bound · *proved*)**.** *Every smooth finite-energy configuration with these limits satisfies $E \geq |W(\phi_+) - W(\phi_-)|$. In the sector $\phi_- = -v$, $\phi_+ = v$, equality is attained by*

$$\phi_{\mathrm{kink}}(x) = v\tanh\big(\mu v(x - x_0)\big), \qquad x_0 \in \mathbb{R}, \qquad E_{\mathrm{kink}} = \frac{4\mu v^3}{3}.$$

Its charge $Q = (\phi_+ - \phi_-)/(2v)$ is +1. The decreasing antikink has charge -1 and the same energy.

Proof. For either sign $s \in \{+1, -1\}$, completing the square gives

$$E = \frac{1}{2}\int_{\mathbb{R}} (\phi' - sW'(\phi))^2 \mathrm{d}x + s\big(W(\phi_+) - W(\phi_-)\big).$$

The last term follows from the chain rule on a finite interval and then the stated limits. Choose s so its product with this difference is nonnegative. The square is nonnegative, proving the bound. Equality requires $\phi' = sW'(\phi)$, the *Bogomolny equation*.

For the increasing sector, this equation is $\phi' = \mu(v^2 - \phi^2)$. Separating variables on $-v < \phi < v$ gives $\operatorname{artanh}(\phi/v) = \mu v(x - x_0)$, hence the displayed profile. Its derivative

is $\mu v^2 \operatorname{sech}^2(\mu v(x - x_0)) = W'(\phi)$. It has the required endpoint limits, so it saturates the bound. Direct endpoint evaluation gives $W(v) - W(-v) = 4\mu v^3/3$. Differentiating the first-order equation gives $\phi'' = W''W' = V'(\phi)$, the static field equation. Thus this saturating configuration is a solution, not only a trial field. Changing its sign gives the antikink and reverses the ordered vacuum pair and charge. ∎

The term *Bogomolny bound* means an energy bound fixed by boundary topological data. Equal endpoint vacua give charge zero. The bound then permits the vacuum configuration. The square completion proves a global energy minimum in each kink sector. It does not imply that every configuration in that sector is stationary.

27.3 Derrick's scaling obstruction

For a smooth linear-target field $\phi\colon \mathbb{R}^d \to \mathbb{R}^N$, use

$$E = E_2 + E_0, \qquad E_2 = \frac{1}{2}\int_{\mathbb{R}^d} |\nabla\phi|^2 \mathrm{d}^d x, \qquad E_0 = \int_{\mathbb{R}^d} V(\phi)\mathrm{d}^d x, \qquad V \geq 0.$$

The subscript counts derivatives in the energy term. A spatial dilation is $\phi_\lambda(x) = \phi(\lambda x)$ for $\lambda > 0$. The chain rule contributes λ^2 to the gradient term. The change of variables contributes λ^{-d} to both terms. Thus

$$E(\lambda) = \lambda^{2-d}E_2 + \lambda^{-d}E_0, \qquad E'(1) = (2-d)E_2 - dE_0.$$

The figure shows this expression with $E_2 = E_0 = 1$.

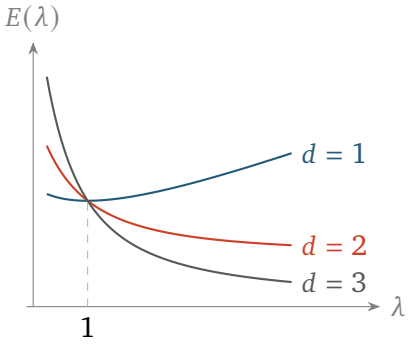


Figure 27.3: Dilation energy for the stated linear-target scalar model. Only the one-dimensional curve has a stationary point for positive E_2 and E_0.

Proposition 27.3 (Derrick's theorem · *proved*). *Let $V\colon \mathbb{R}^N \to [0,\infty)$ be smooth. Every smooth finite-energy static solution $\Delta\phi = \nabla V(\phi)$ of this model on $\mathbb{R}^d$, for $d \geq 2$, is a constant vacuum.*

Proof. **Step 1: the stress identity.** First justify the stationary scaling identity without assuming a variation with noncompact support is admissible. Use the Euclidean metric to raise indices and let $e = |\nabla\phi|^2/2 + V(\phi)$. The stress tensor and its divergence are

$$T^i{}_j = \partial^i\phi \cdot \partial_j\phi - \delta^i_j e, \qquad \partial_i T^i{}_j = (\Delta\phi - \nabla V(\phi)) \cdot \partial_j\phi = 0.$$

To see the cancellation, differentiate the first term by the product rule. Its second-derivative term $\partial^i\phi \cdot \partial_i\partial_j\phi$ is the derivative of $|\nabla\phi|^2/2$ and cancels that part of $\partial_j e$. The remaining potential derivative is $\nabla V \cdot \partial_j\phi$.

Step 2: the cutoff limit. Fix one smooth bump χ from Lecture 21, equal to one on the unit ball and zero outside the radius-two ball. Put $\chi_R(x) = \chi(x/R)$. The chain rule gives $|\nabla\chi_R| \le C/R$ with $C = \sup|\nabla\chi|$. Integrate the divergence of $T^i{}_j x^j \chi_R$ and use compact support to obtain

$$\int_{\mathbb{R}^d} \chi_R T^i{}_i \mathrm{d}^d x = -\int_{\mathbb{R}^d} T^i{}_j x^j \partial_i \chi_R \mathrm{d}^d x.$$

The absolute value of the right side is bounded by a constant times $\int_{R<|x|<2R}(|\nabla\phi|^2 + V)\mathrm{d}^d x$, which tends to zero by finite energy. The trace is integrable, so dominated convergence on the left gives $(2-d)E_2 - dE_0 = 0$. For $d > 2$, nonnegativity forces $E_2 = E_0 = 0$, hence a constant vacuum.

Step 3: dimension two. For $d = 2$ the identity gives $E_0 = 0$. Continuity and nonnegativity imply $V(\phi(x)) = 0$ everywhere. Each such field value minimizes V, so $\nabla V(\phi(x)) = 0$. The field equation becomes $\Delta\phi = 0$. Let u be any component of a derivative $\partial_i\phi$. It is harmonic and square-integrable. Its spherical mean about any point p is independent of radius: its derivative is the normalized outward flux of ∇u, zero by Stokes and $\Delta u = 0$. The mean tends to $u(p)$ at radius zero, so averaging the sphere means radially gives the ball mean. Cauchy–Schwarz therefore yields

$$|u(p)| = \left|\frac{1}{|B_R|}\int_{B_R(p)} u\,\mathrm{d}^d x\right| \le |B_R|^{-1/2}\|u\|_{L^2(\mathbb{R}^d)} \longrightarrow 0.$$

Thus all derivatives vanish and ϕ is a constant vacuum in this case too. ∎

A constrained sphere-valued sigma model is outside the linear-target hypothesis: its field equation is not $\Delta\phi = \nabla V$ for a potential on all of $\mathbb{R}^N$. Gauge fields and higher-derivative energies also have different scaling powers. They need not satisfy this obstruction. Our dilation is $\phi(\lambda x)$, as in Manton–Sutcliffe, equations (4.17)–(4.19), pp. 84–85. Inverting the scale parameter changes the displayed derivative sign but not the obstruction.

27.4 The abelian Higgs vortex and its flux

A *Higgs field* here is a charged complex scalar $\phi\colon \mathbb{R}^2 \to \mathbb{C}$. Let $q \neq 0$ be its signed charge and put $a = q/\hbar$. Use a smooth real potential A_{em} on the plane and Lecture 21's derivative $D = \mathrm{d} - \mathrm{i}aA_{\mathrm{em}}$. Write $A_{\mathrm{em}} = A_x\mathrm{d}x + A_y\mathrm{d}y$ and $B = \partial_x A_y - \partial_y A_x$. For $v > 0$ and $\lambda > 0$, the static abelian Higgs energy is

$$E = \int_{\mathbb{R}^2}\left[\frac{1}{2}B^2 + \frac{1}{2}\left(|D_x\phi|^2 + |D_y\phi|^2\right) + \frac{\lambda}{8}\left(|\phi|^2 - v^2\right)^2\right]\mathrm{d}x\mathrm{d}y.$$

Its vacuum field values have modulus v, a circle. Write $r = \sqrt{x^2+y^2}$. For $r = R$, write $\phi_R(\theta) = \phi(R\cos\theta, R\sin\theta)$. The symbol D_θ means the covariant derivative along the angular parameter, so $D_\theta\phi_R = \partial_\theta\phi_R - \mathrm{i}a(A_{\mathrm{em}})_\theta\phi_R$. It includes the radius factor relative to a unit tangent derivative.

Theorem 27.4 (Vortex flux · *sketched; flux proved*)**.** *Assume B is integrable, $|\phi_R| \to v$ uniformly, and $\int_0^{2\pi}|D_\theta\phi_R|\mathrm{d}\theta \to 0$ as $R \to \infty$. Let n be the winding of $\phi_R/|\phi_R|$ for large R. The total flux is*

$$\Phi = \int_{\mathbb{R}^2} B\mathrm{d}x\mathrm{d}y = \frac{2\pi n}{a} = \frac{2\pi\hbar n}{q}.$$

At critical coupling $\lambda = a^2$, finite-energy fields in this sector obey $E \ge \pi v^2|n|$.

Flux proof. The amplitude is bounded below by $v/2$ outside a sufficiently large disk. Thus the nonzero fields on any two large circles are homotopic through the intervening annulus, and their winding n agrees. On each circle, the imaginary part of the logarithmic derivative gives

$$a \oint_{r=R} A_{\mathrm{em}} = 2\pi n - \int_0^{2\pi} \operatorname{Im}\left(\frac{D_\theta \phi_R}{\phi_R}\right) \mathrm{d}\theta.$$

The last integral tends to zero by the amplitude lower bound and the assumed angular derivative integral. Stokes on the disk gives $\oint A_{\mathrm{em}} = \int_{r\le R} B \mathrm{d}x\mathrm{d}y$. Integrability of B gives the total flux in the limit, proving the formula. ∎

These asymptotic conditions are part of the sector. Finite energy alone was not used to infer pointwise limits. The magnetic flux now has the same integer origin as Lecture 3's winding and Lecture 22's quantization. Gauge maps in this discussion are smooth on the whole plane. Their restriction to a circle contracts through the disk to the value at its centre. Its winding is therefore zero, so these gauge changes preserve n.

To sketch the bound, let $j_i = \operatorname{Im}(\bar{\phi} D_i \phi)$, for $i \in \{x, y\}$, be the components of a real boundary-current one-form, distinct from the physical current of Lecture 15. Set $\operatorname{curl} j = \partial_x j_y - \partial_y j_x$. On a radius-$R$ circle its circulation satisfies

$$\left| \oint_{r=R} j \right| \le \sup_\theta |\phi_R| \int_0^{2\pi} |D_\theta \phi_R| \mathrm{d}\theta \longrightarrow 0.$$

The sector assumptions therefore remove this boundary term. Using $[D_x, D_y]\phi = -\mathrm{i}aB\phi$ gives $\operatorname{curl} j = 2\operatorname{Im}(\overline{D_x \phi} D_y \phi) - aB|\phi|^2$. For either sign $s \in \{+1, -1\}$, the resulting square completion at $\lambda = a^2$ is

$$E = \frac{1}{2} \int_{\mathbb{R}^2} \left[|(D_x + \mathrm{i}sD_y)\phi|^2 + \left(B + \frac{sa}{2}(|\phi|^2 - v^2) \right)^2 \right] \mathrm{d}x\mathrm{d}y + \frac{sav^2}{2}\Phi.$$

Obtain this identity first on disks using Stokes for the curl term. The boundary-current circulation tends to zero by the estimate above. Finite energy and integrability of the magnetic field give the limit. Choosing $s = \operatorname{sgn}(n)$ for $n \neq 0$ gives the bound. Solutions of the two first-order equations obtained by setting both squares to zero saturate the bound. Their existence requires an analytic input. Away from the zeros of ϕ, these first-order equations reduce to one nonlinear elliptic equation for $\log|\phi|^2$. The prescribed zeros supply its point-source conditions. Solving that boundary problem constructs vortices. The analytic existence step is cited from Manton–Sutcliffe, §7.8, pp. 199–200. Their normalization is $a = v = 1$, with critical $\lambda = 1$ and bound $\pi|n|$. In general their connection potential is aA_{em}. This potential rescaling alone does not equate their unit-coupling energy with the general-a energy printed here.

The figure shows a positive winding with $a > 0$, fixing the circulation arrows.

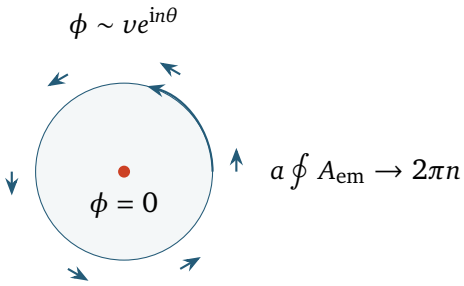


Figure 27.4: The asymptotic Higgs phase winds counterclockwise. For $a > 0$ and $n = 1$, its covariantly constant limit requires positive potential circulation and flux.

Example 27.5 (From asymptotic phase to flux)**.** Assume $\phi_R \to ve^{in\theta}$ with its first angular derivative, and $D_\theta \phi_R \to 0$ uniformly. The limiting derivative equation then gives $(A_{\mathrm{em}})_\theta \to n/a$ in this gauge. With B integrable, the flux proof needs only the integrated angular condition, and gives $\Phi = 2\pi n/a$. A negative signed charge reverses the magnetic flux for the same positive winding.

For a concrete smooth configuration of winding one, use

$$\phi(x,y) = \frac{v(x+\mathrm{i}y)}{\sqrt{1+x^2+y^2}}, \qquad A_{\mathrm{em}} = \frac{-y\mathrm{d}x + x\mathrm{d}y}{a(1+x^2+y^2)}, \qquad B = \frac{2}{a(1+x^2+y^2)^2}.$$

It has a Higgs zero at the origin and no singular potential there. On a radius-R circle, $\phi_R = vRe^{\mathrm{i}\theta}/\sqrt{1+R^2}$ and $(A_{\mathrm{em}})_\theta = R^2/[a(1+R^2)]$. Thus $D_\theta \phi_R = \mathrm{i}\phi_R/(1+R^2)$ tends uniformly to zero. The flux is

$$\Phi = \frac{4\pi}{a}\int_0^\infty \frac{r\mathrm{d}r}{(1+r^2)^2} = \frac{2\pi}{a}.$$

Direct differentiation of the displayed fields gives

$$D_x\phi = \frac{v}{(1+r^2)^{3/2}}, \qquad D_y\phi = \frac{\mathrm{i}v}{(1+r^2)^{3/2}}.$$

Thus the sum of the squared norms of these two derivatives is $2v^2/(1+r^2)^3$. Substitution of $t = 1+r^2$ makes the radial integrals of $r/(1+r^2)^p$ equal to $1/[2(p-1)]$, for $p \in \{2,3,4\}$. Each energy density term is integrable. The total energy is

$$E = \pi\left[\frac{2}{3a^2} + \frac{v^2}{2} + \frac{\lambda v^4}{8}\right].$$

This field demonstrates the finite-energy flux sector. It is a trial configuration, not an asserted solution of the vortex field equations.

27.5 The monopole charge

Remark 27.6 (The Yang–Mills–Higgs monopole)**.** In the SU(2) Yang–Mills–Higgs model, an adjoint Higgs field is a vector in the Lie algebra that transforms by conjugation. Its fixed-norm vacuum directions form $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$. With a continuous nonzero Higgs limit at infinity, the normalized field defines a map from the outward spatial sphere to this vacuum sphere. Its degree is the topological monopole charge $Q \in \mathbb{Z}$, as in Lecture 26. Orient the Lie algebra by Lecture 23's basis T^1, T^2, T^3. The hedgehog with coefficients x^a/r in that basis restricts to the identity direction map between outward spheres, so it has $Q = +1$. The

remaining U(1) connection acts on charged line bundles at infinity. Lecture 22 then requires $q\Phi_{\mathrm{m}} = 2\pi\hbar n$ for each such charge convention. Relating that integer to the degree also requires specifying the chosen charged representation. The full 't Hooft–Polyakov ansatz and energy bound are on Sheet 27 (Manton–Sutcliffe, §8.2).

Self-check. For the scalar kink model, list the possible values of $Q = (\phi_+ - \phi_-)/(2v)$ for vacuum endpoints.

Answer. The values are +1 for $(-v, +v)$, −1 for $(+v, -v)$ and zero for equal endpoints. The endpoint pair is ordered.

Summary. Continuous vacuum boundary data define sectors that an evolution with jointly continuous vacuum boundary data cannot change. Finite energy alone does not supply those limits. The scalar kink saturates $E \geq |\Delta W|$ and has signed charge +1. Derrick's theorem excludes nonconstant smooth finite-energy static solutions of the stated nonnegative-potential linear-target scalar model for $d \geq 2$. The gauge vortex lies outside that model and has flux $2\pi\hbar n/q$ under the stated asymptotic conditions. Its critical-coupling bound follows by square completion. Existence uses the cited analytic input. Lecture 28 uses a quantized sphere flux to count chiral quantum zero modes.

Reading. Manton–Sutcliffe, §4.2, pp. 84–85; §5.1–5.2, pp. 110–113; §7.3, pp. 164–165; §7.8, pp. 197–200; §8.2.